

Small Gift Today





SINICA × NCHC
07 AUG 2026 | TAINAN, TAIWAN
DESIGN DRAFT | SAMPLE 01 / 10

THANK YOU FOR ENGAGING

CUDA-Q
AGENTIC CODING BOOTCAMP

ACTIVE PARTICIPATION RECOGNITION

For your curiosity, thoughtful questions, and active participation in exploring CUDA-Q, quantum-GPU computing, and agentic coding.

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國家高速網路與計算中心

National Center for High-performance Computing

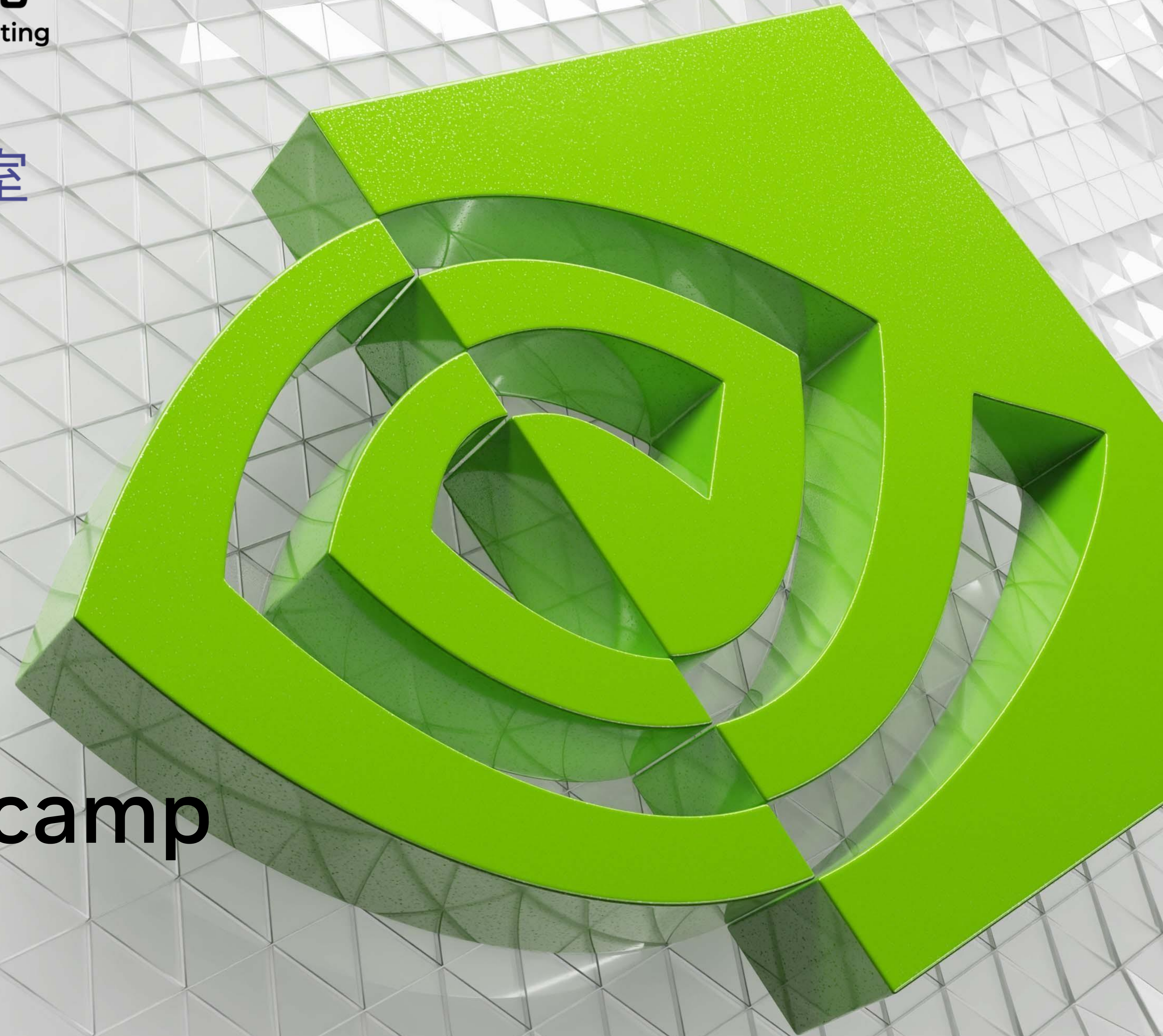


量子電子元件實驗室

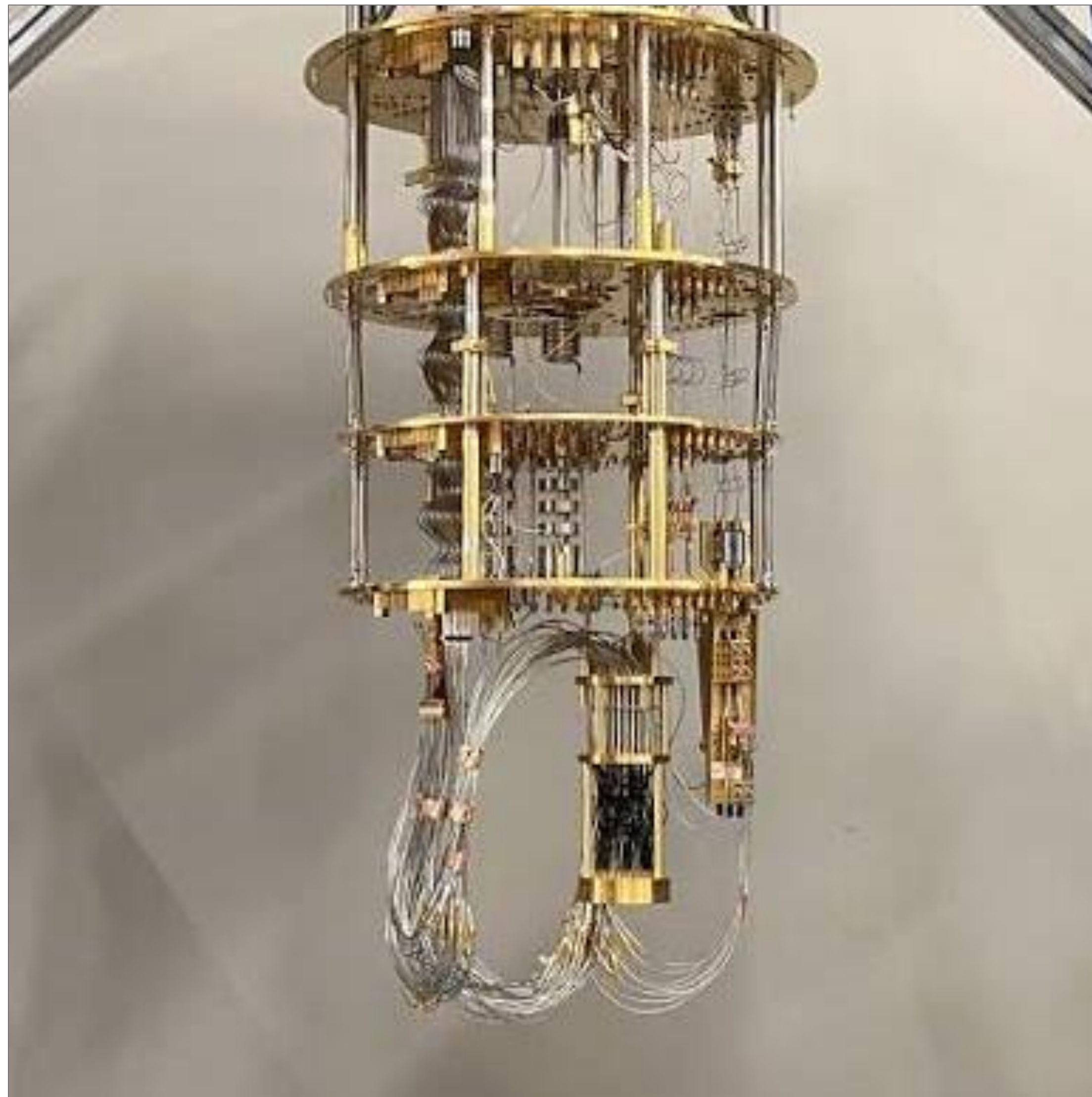
Quantum Electronics Laboratory

SINICA x NCHC x NVIDIA CUDA-Q Agentic Coding Bootcamp

2026



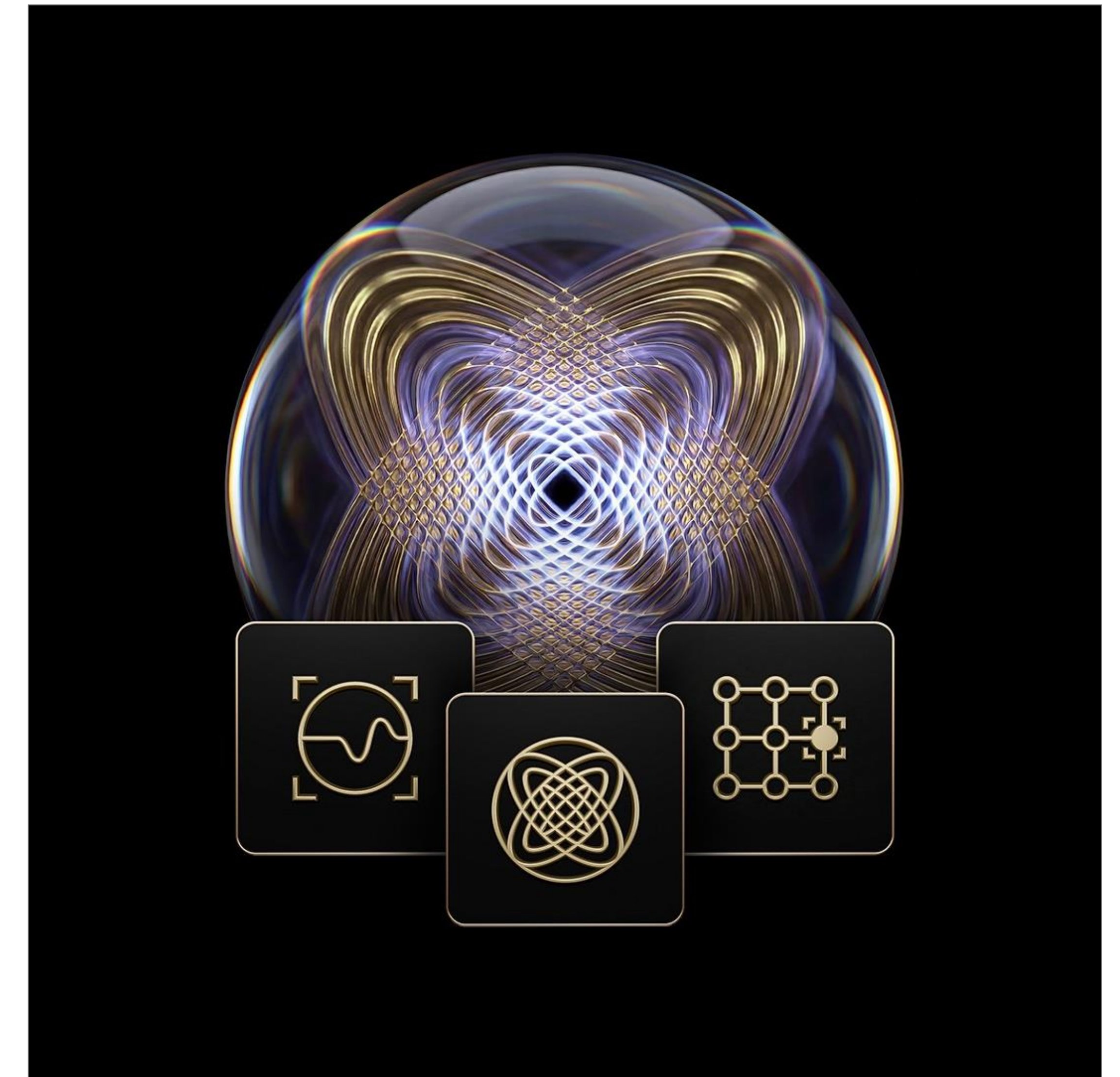
Bootcamp Highlights



Academia Sinica QPU system
and agentic calibration



NCHC Nano4 H200s with OpenStack
and research sharing



NVIDIA CUDA-Q + NVIDIA Ising Model +
Nemotron3 customized materials

Team Roster

More than 30 people contribute to the event! Many thanks.



量子電子元件實驗室
Quantum Electronics Laboratory



陳啟東
柯忠廷
呂德輝
張淮竣
Roxy Liu
IT Team

[Quantum Team]
李泰岳, 何明鋼, 劉沛豪, 姚欣妤, 俞昊天, 楊安正

[RAP LLM API Team]
周志豪, 許伯豪, 林昀德

[OpenStack H200 VMs Team]
陳威宇, 吳佳諺, 蔡育欽, 潘怡倫, 郭芳安

[Joint Lab]
吳宙錦

Pika Wang
Jiun-Cheng Jiang
Jay Chen
Eric Kang
Vincent Chiu
OpenHackathon.org Team
NVIDIA CUDA-Q team

SINICA NCHC CUDA-Q AGENTIC CODING BOOTCAMP

Hands-on training on NVIDIA Quantum Platform, hybrid quantum-classical workflows, and agentic AI for real quantum hardware



2026/8/7 (五)
9:30-16:20



中央研究院南院
關鍵議題研究中心1F演講廳



關鍵議題研究中心
Research Center for Critical Issues

09:30-10:00 Registration & Setup

10:00-10:15 Opening

Academia Sinica |
陳啟東教授

10:15-10:45 NVIDIA Quantum Platform
Overview: CUDA-Q & NVIDIA Ising

NVIDIA | Pika Wang

10:45-11:15 NVIDIA CUDA-Q Environment Setup

11:15-12:00 Basic CUDA-Q Hands-on + Ising
Calibration Playground (NIM)

12:00-13:00 Lunch Break



13:00-14:00 Superconducting QPU
Introduction & Agentic Calibration

Academia Sinica |
陳啟東教授
NCHC | 李泰岳博士,
何明鋼博士

14:00-14:10 NCHC Tech Sharing

NCHC | 李泰岳博士

14:10-14:20 Agentic Coding with NCHC RAP
Service

NVIDIA | Jay Chen

14:20-14:40 Tea Break



14:40-15:20 Advanced CUDA-Q Hands-on with
RAP

NVIDIA | Pika Wang

15:20-16:00 Bring-Your-Own-Code with CUDA-
Q Agentic Programming

16:00-16:20 Q&A + Open Discussion



SINICA x NCHC x NVIDIA Workshops/Bootcamps/Hackathons

Since 2020

Events
56+

Developers
2800+



Bootcamps are published on NCHC Youtube Channel

智慧未來
先知先行
NCHC • AI Enabler



國網中心NCHC

@nchc3005 • 1.97K subscribers • 213 videos

國家高速網路與計算中心...more

nchc.org.tw and 2 more links

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育訓練_Taiwan AI RAP

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育訓練 Day2

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SEAIP 2023 & International HPC Symposium
event video

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長訪談

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加速運算 Day1

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Workshops and Bootcamps are Open-Sourced

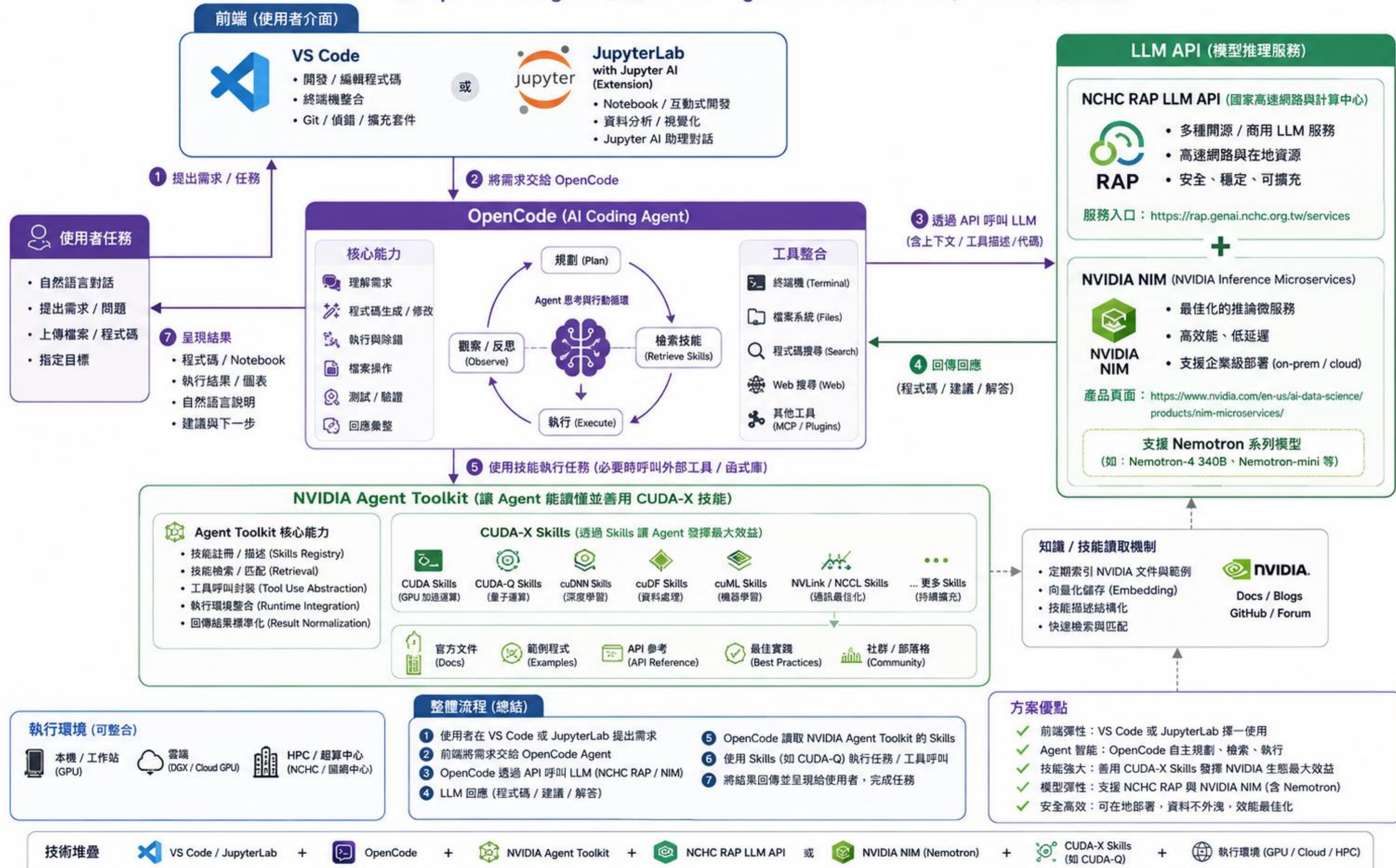
<https://github.com/nqobu/nvidia>

Repository for NCHC-NVIDIA Joint Lab

- tutorials - NVIDIA online courses/tutorials in AI/HPC
- 20210412 - NVIDIA Techniques Sharing 2021
- 20210706 - NCHC Techniques Sharing 2021
- 20211202 - NVIDIA Techniques Update 2021
- 20211221 - AI+HPC: 利用 NVIDIA Modulus 實踐 PINN 於物理模擬
- 20220415 - NVIDIA Techniques Update 2022
- 20220530 - NCHC-NVIDIA Techniques Sharing 2022
- 20220629 - NCHC PINN 與 NVIDIA Modulus 實作訓練營
- 20221111 - NCHC Quantum Computing Workshop / 量子計算模擬
- 20230413 - NVIDIA Techniques Sharing 2023
- 20230517 - NVIDIA Techniques Briefing: NVIDIA Federated Learning
- 20230525 - AI for Science: NVIDIA Modulus 及 NVIDIA Omniverse 實作
- 20230727 - NCHC N-Way to GPU Programming Bootcamp / 多 GPU 程式設計訓練課程
- 20230821 - NVIDIA Techniques Salon 2023: Programming the NVIDIA Superchip
- 20231207 - NCHC Open Hackathon 2023
- 20240410 - NCHC Quantum Computing Bootcamp 2024 / NVIDIA CUDA-Q and cuQuantum
- 20240506 - AI for Science: NVIDIA Modulus, NVIDIA Omniverse, and NVIDIA Earth-2
- 20240508 - NCHC Techniques Sharing 2024
- 20240626 - NCHC AI for Science Bootcamp 2024 / NVIDIA Modulus 物理模擬計算
- 20240806 - NCHC End-to-end LLM Bootcamp 2024 / NVIDIA NeMo 大型語言模型框架
- 20240924 - NCHC N-Way Bootcamp 2024 / NVIDIA GPU 加速運算
- 20241114 - NVIDIA Earth-2 Overview
- 20241129 - NCHC x NTU - NVIDIA BioNeMo Protein Design Workshop 2024
- 20241204 - NCHC Open Hackathons 2024
- 20250218 - NCHC Grace Workshop 2025
- 20250415 - NCHC N-Way Bootcamp 2025 / NVIDIA GPU 加速運算訓練課程
- 20250617 - NCHC LLM Bootcamp 2025 / 大語言模型訓練課程
- 20250723 - NCHC CUDA-Q Bootcamp 2025 / 量子計算模擬訓練課程
- 20250904 - NCHC x NCKU - Digital Biology Bootcamp / 數位生物技術訓練營
- 20251110 - NCHC Quantum-HPC Tech Salon
- 20251112 - NCHC Open Hackathons 2025
- 20260310 - NCHC AI Powered Physics Bootcamp, 2026
- 20260520 - NCHC LLM & Agentic AI Bootcamp, 2026

OpenCode + VSCode / JupyterLab + NCHC RAP LLM API / NVIDIA NIM + Nemotron

讓 OpenCode Agent 善用 NVIDIA Agent Toolkit 的 Skills (CUDA-X) 完成任務

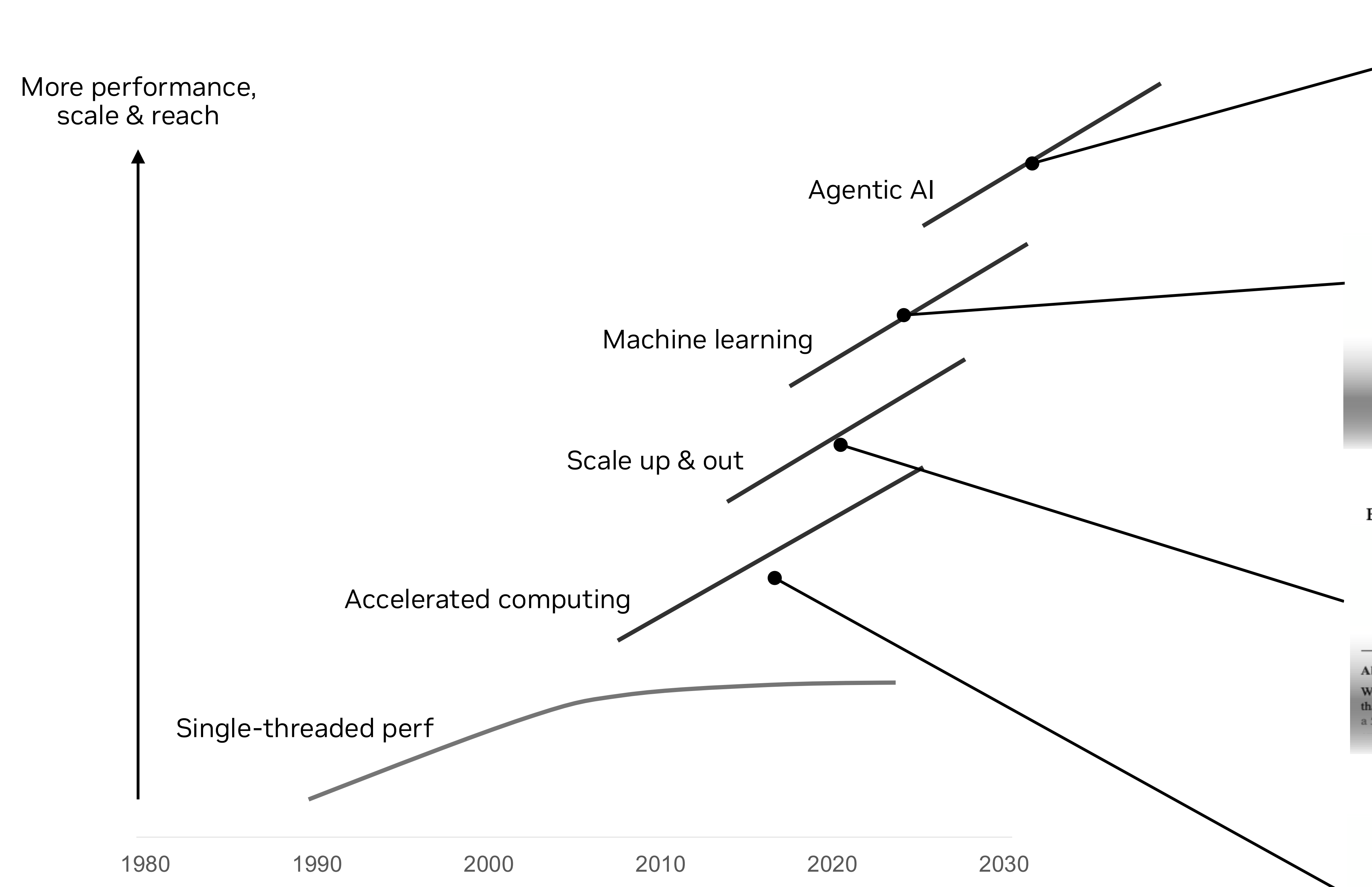


Agentic AI for Science



Incredible Growth in Scale and Reach of Scientific Computing

Expanding capabilities, accessibility and applications leading to new discoveries



Increasing capabilities and access for scientific computing

OpenFold: Retraining AlphaFold2 yields new insights into its learning mechanisms and capacity for generalization

Gustaf Ahdriz^{*1,2}, Nazim Bouatta^{*3}, Christina Floristean¹, Sachin Kadyan¹, Qinghui Xia¹, William Gerecke³, Timothy J O'Donnell⁴, Daniel Berenberg^{5,6}, Ian Fisk⁷, Niccolò Zanichelli⁸, Bo Zhang⁹, Arkadiusz Nowaczynski¹⁰, Bei Wang¹⁰, Marta M Stepniewska-Dziubinska¹⁰, Shang Zhang¹⁰, Adegoke Ojewole¹⁰, Murat Efe Guney¹⁰, Stella Biderman^{11,12}, Andrew M Watkins⁵, Stephen Ra⁵, Pablo Ribalta Lorenzo¹⁰, Lucas Nivon¹³, Brian Weitzner¹⁴, Yih-En Andrew Ban¹⁵, Peter K Sorger³, Emad Mostaque¹⁶, Zhao Zhang¹⁷, Richard Bonneau⁵, and Mohammed AlQuraishi¹

Universal Quantum Computer Simulation of 50 Qubits on Europe's First Exascale Supercomputer Harnessing Its Heterogeneous CPU-GPU Architecture

Hans De Raedt^a, Jiri Kraus^b, Andreas Hertel^a, Vrinda Mehta^{a,*}, Mathis Bode^a, Markus Hrywniak^b, Kristel Michiels^a, Thomas Lippert^a

^a Jülich Supercomputing Centre, Forschungszentrum Jülich, D-52425 Jülich, Germany
^b NVIDIA, Würselen, Germany

Abstract

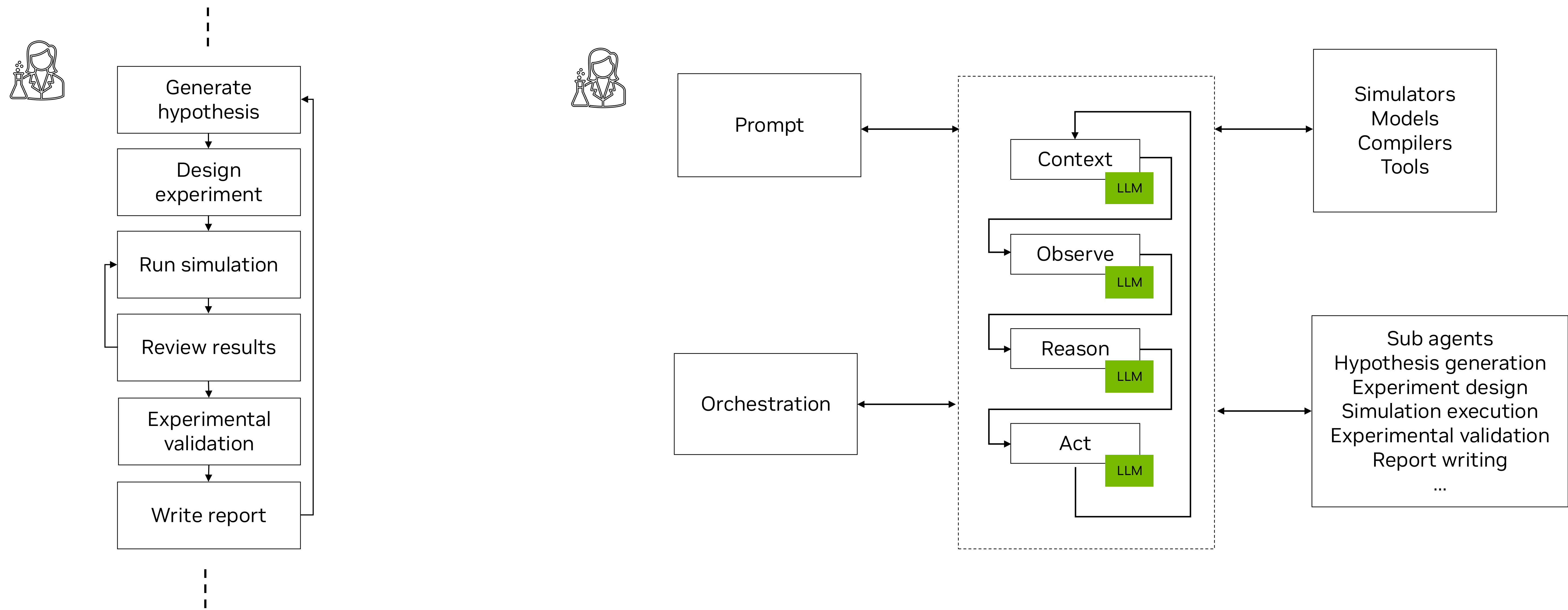
We have developed a new version of the high-performance Jülich universal quantum computer simulator (JUQCS-50) that leverages key features of the GH200 superchips as used in the JUPITER supercomputer, enabling simulations of a 50-qubit universal quantum computer for the first time. JUQCS-50 achieves this through three key innovations: (1)

Breaking the Million-Electron and 1 EFLOP/s Barriers: Biomolecular-Scale *Ab Initio* Molecular Dynamics Using MP2 Potentials

Ryan Stocks School of Computing Australian National University ryan.stocks@anu.edu.au	Jorge L. Galvez Vallejo School of Computing Australian National University jorge.galvezvallejo@anu.edu.au	Fiona C. Y. Yu School of Computing Australian National University fiona.yu@anu.edu.au	Calum Snowdon School of Computing Australian National University calum.snowdon@anu.edu.au
Elise Palethorpe School of Computing Australian National University elise.palethorpe@anu.edu.au	Jakub Kurzak Advanced Micro Devices, Inc. jakub.kurzak@amd.com	Dmytro Bykov Computing for Chemistry Oak Ridge National Lab bykovd@ornl.gov	Giuseppe M. J. Barca School of Computing University of Melbourne giuseppe.barca@unimelb.edu.au

Agentic AI Expands Reach of Scientific Computing

Expanding capabilities, accessibility and applications leading to new discoveries

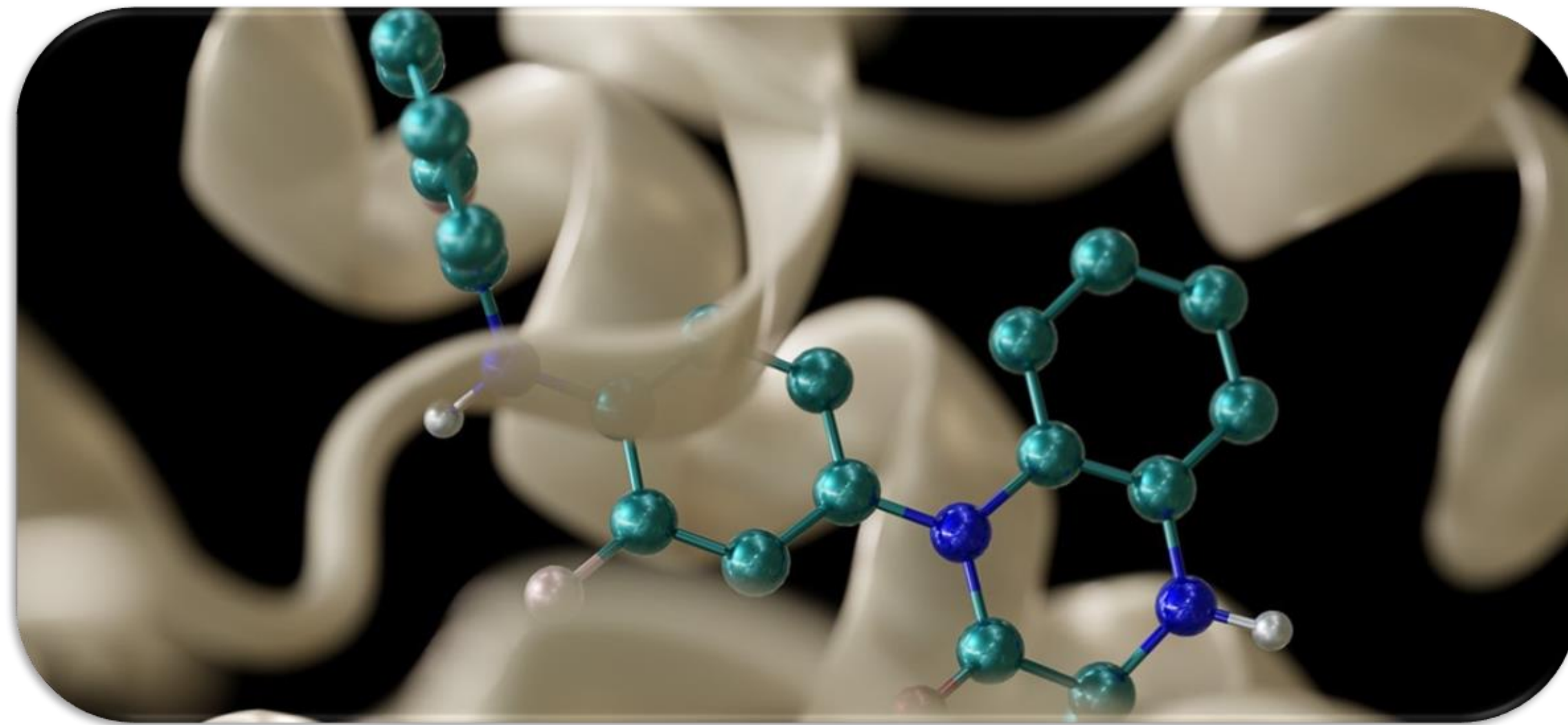


Traditional Workflow

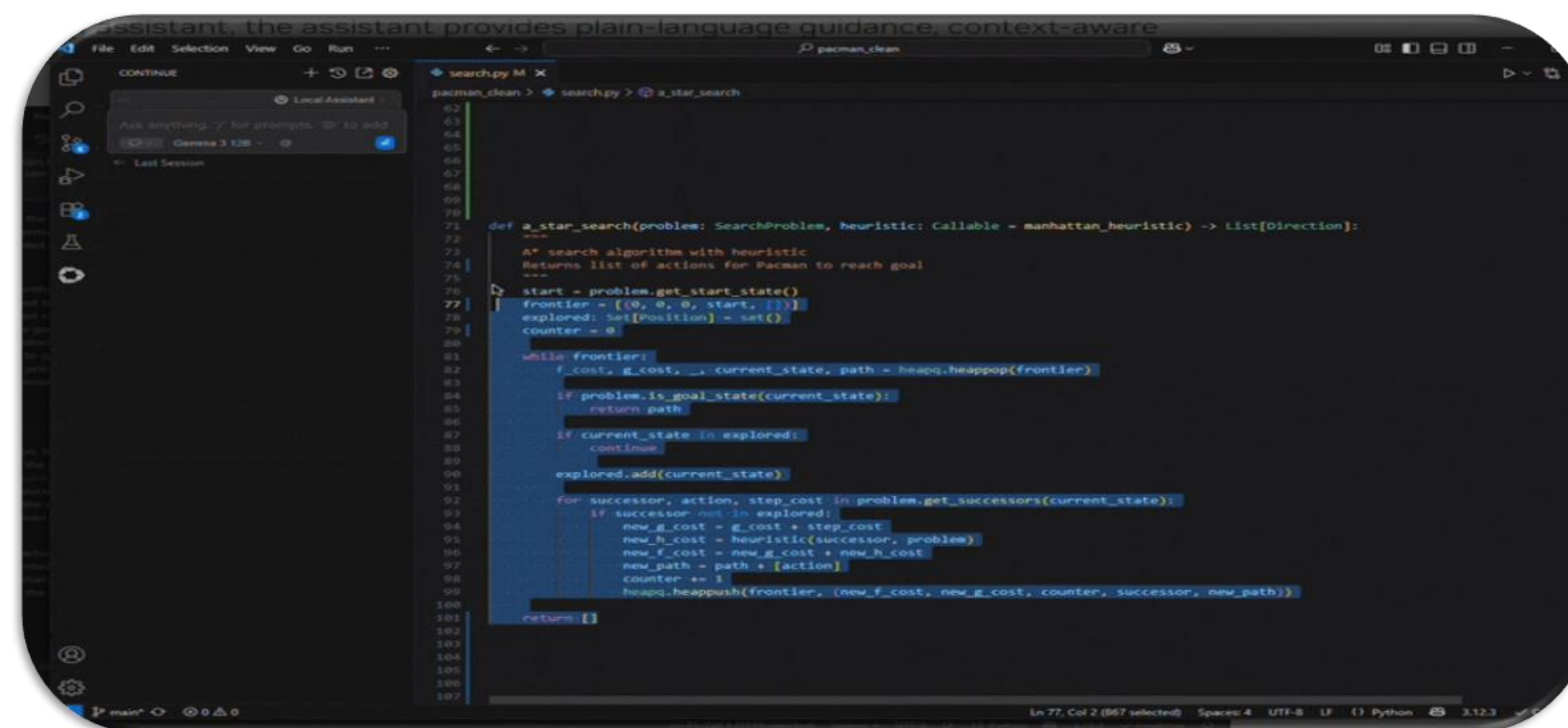
Agentic Workflow

New Trend: AI Scientists

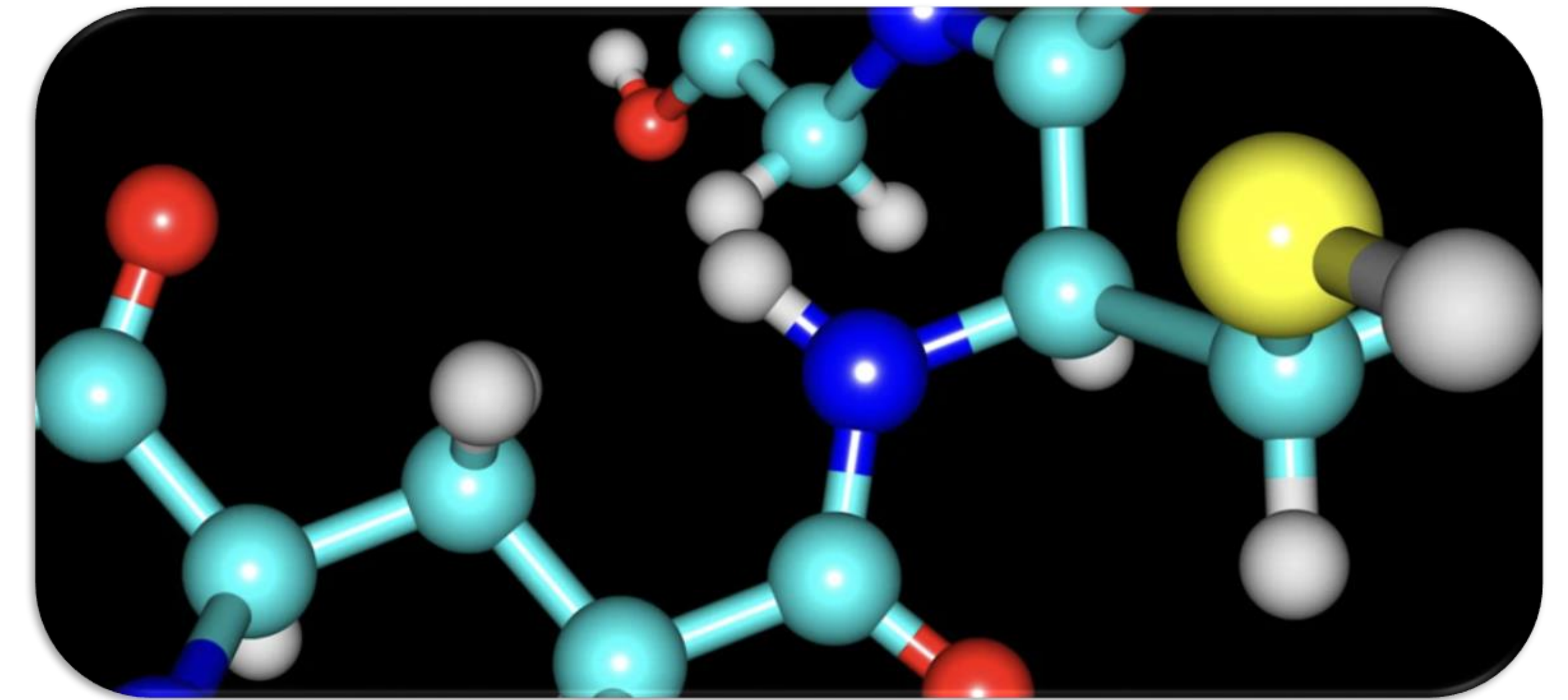
AI Scientists emerging across every domain



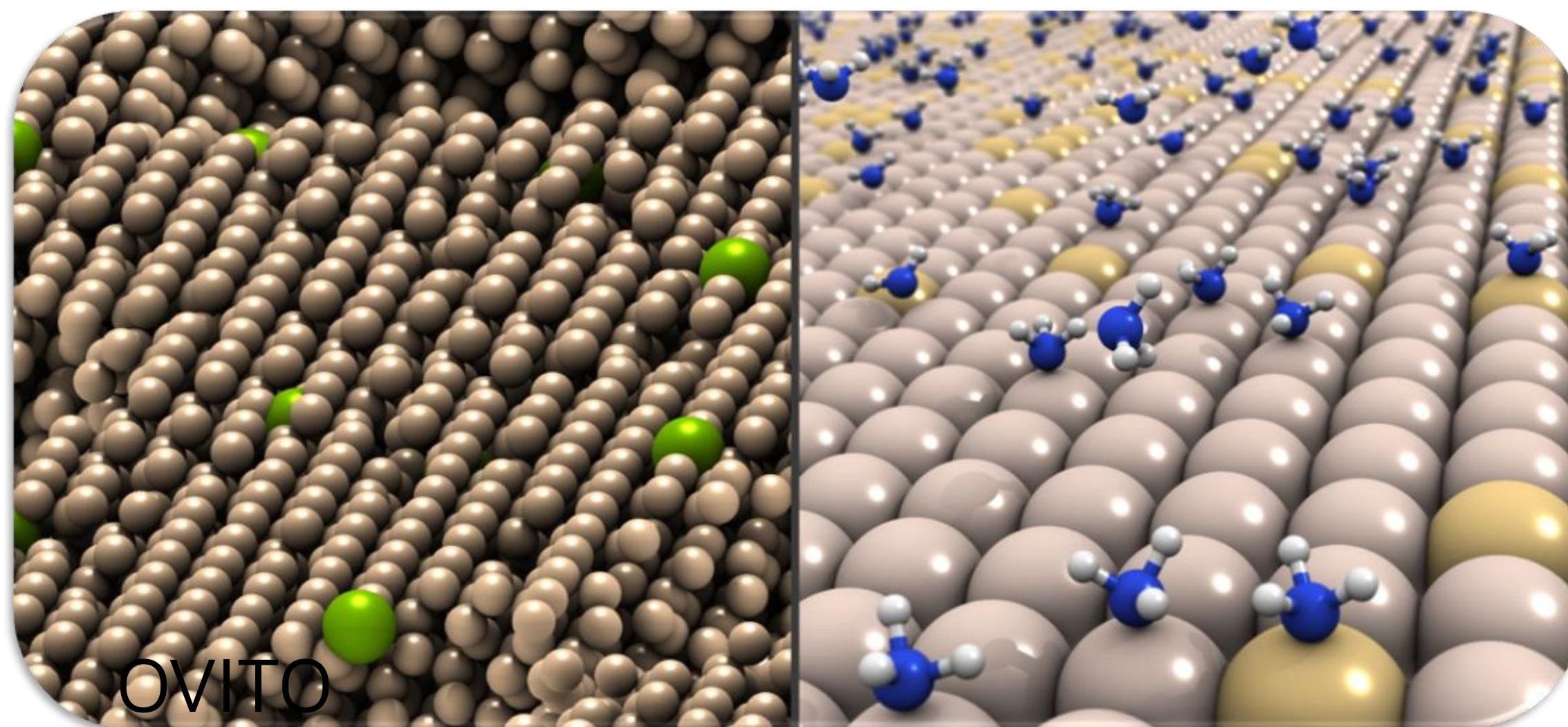
Automating Drug Discovery



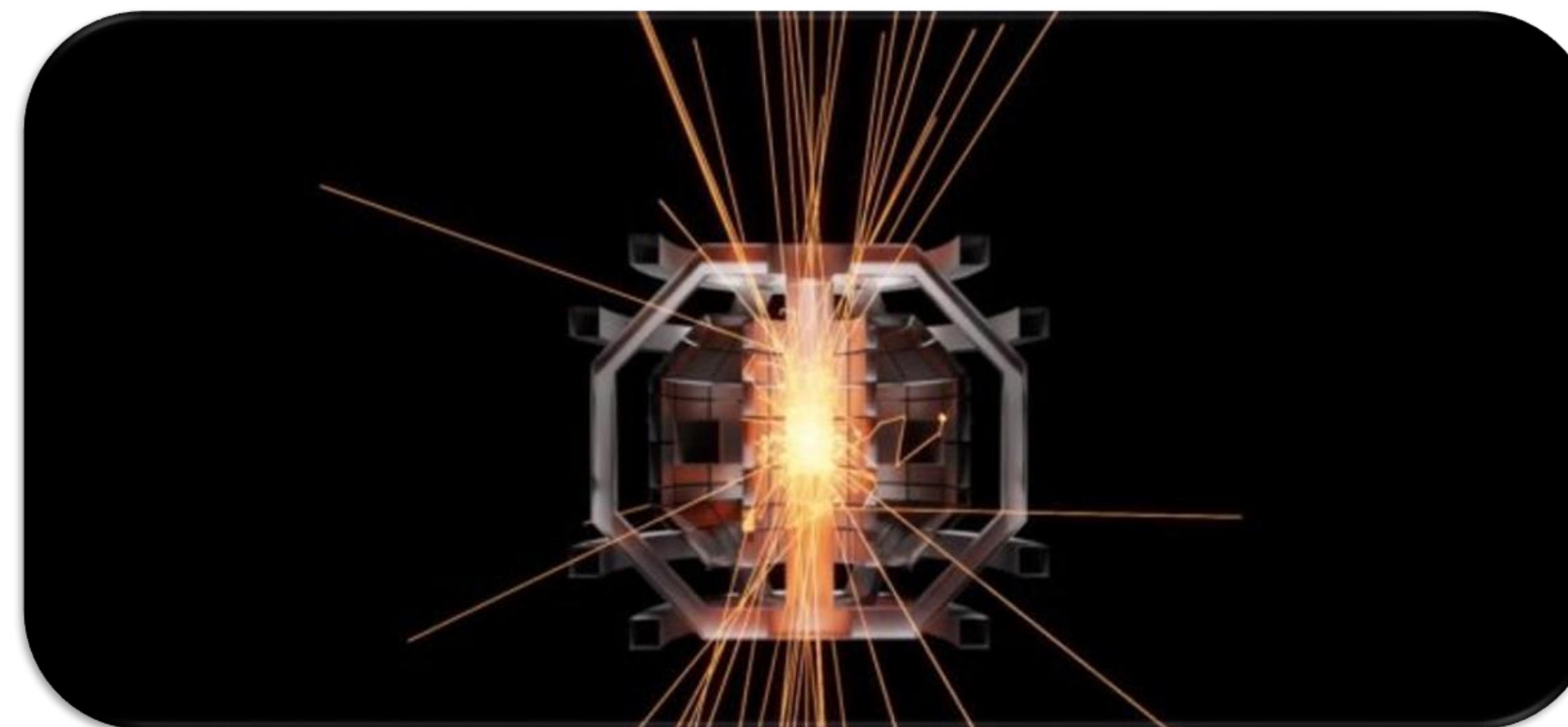
Automating Coding



Designing New Molecules



Materials Candidate Generation



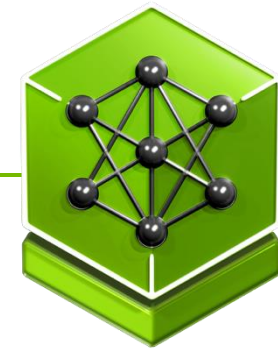
Fusion Hypothesis Generation



Optimizing Clinical Workflows

Nemotron 3 Open Models

Frontier Smart. Faster. More Cost Efficient



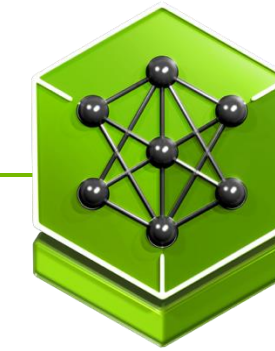
Ultra

550B-A55B

Frontier Smart

5X Faster

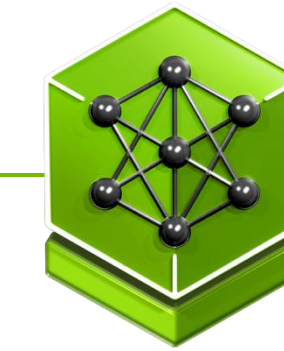
30% Lower Compute Cost



Super

120B-A12B

4X Throughput with
High Accuracy



Nano

30B-A3B

60% Savings with
Thinking Budget

Open Datasets

Pre-Training Tokens, Post-
Training Samples, RL Tasks

NeMo Gym

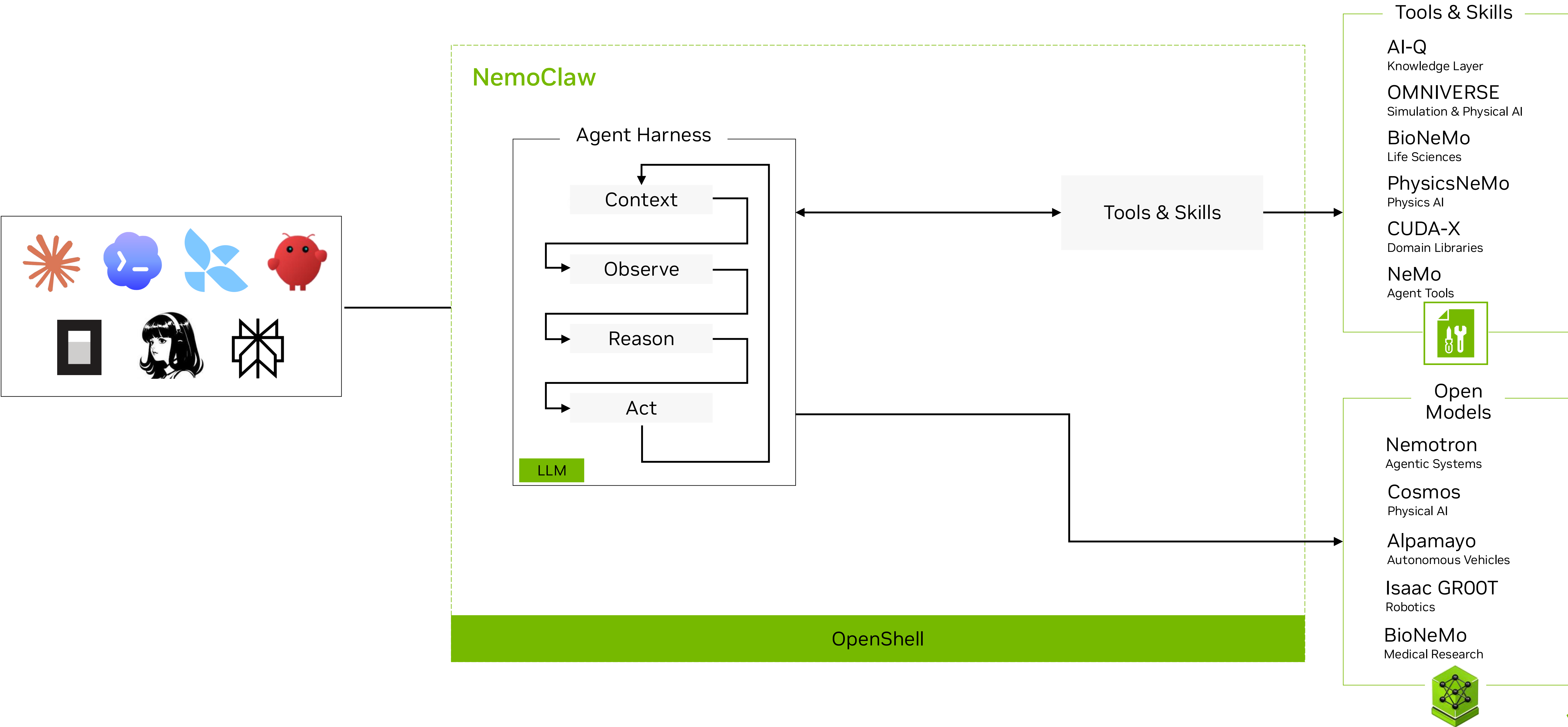
RL Environments & Skills

Open Research

Papers, Samples

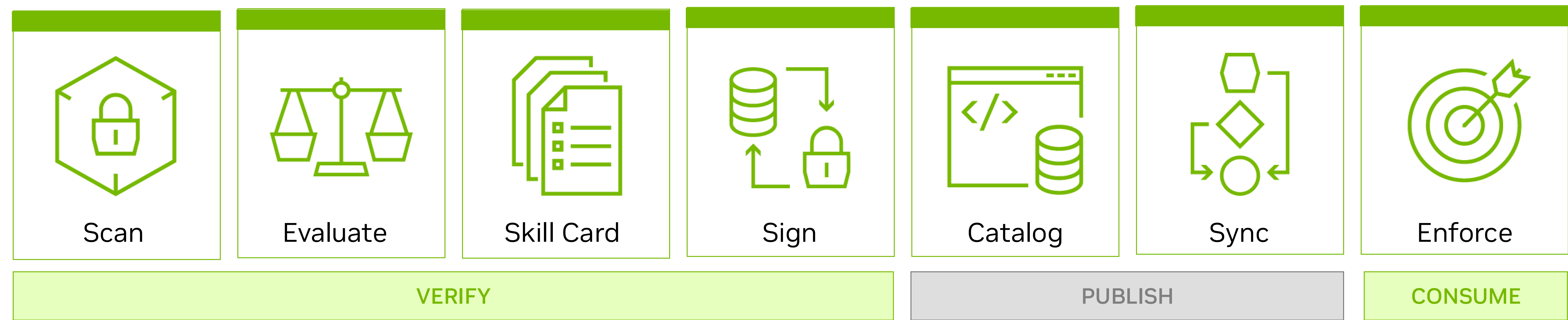
NVIDIA Agent Toolkit for Enterprise AI

Building Blocks for Specialized Agents

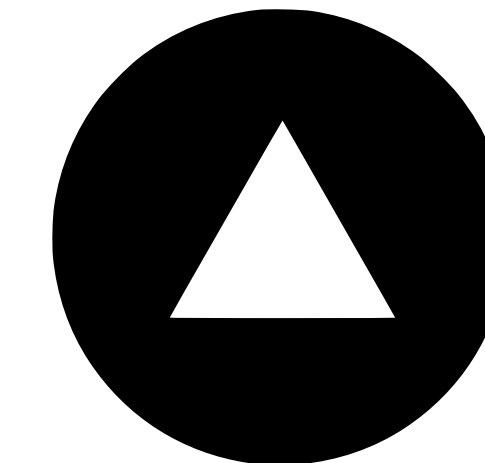


NVIDIA Libraries Available as Skills for Agents

github.com/NVIDIA/skills



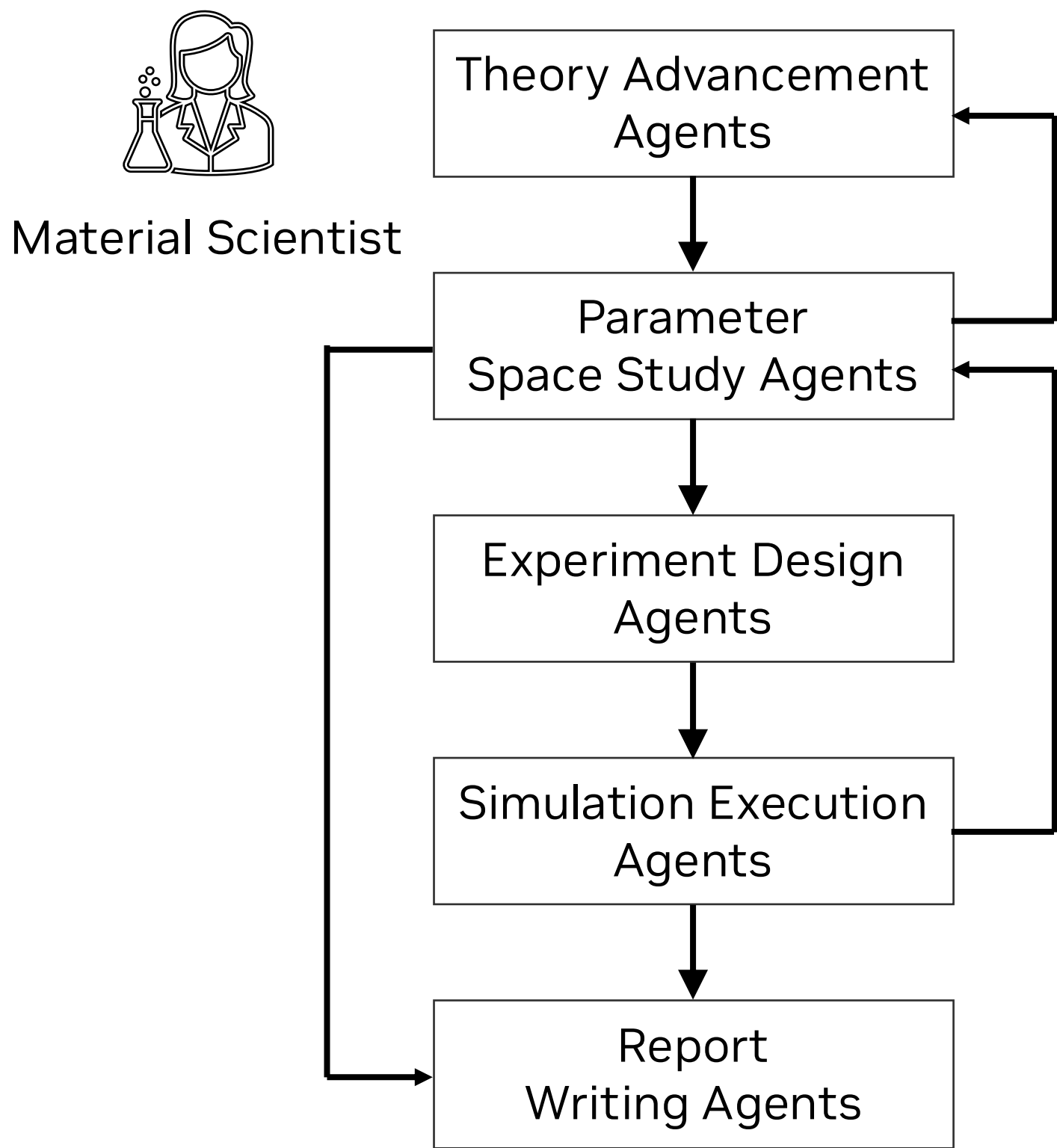
Available as plugins with your preferred agent:



LANL, University of Michigan, and Open AI Demonstrate URSA

- Agentic AI for Material Discovery

AI framework helps scientists plan, run, and analyze experiments connected to scientific foundation model (SciFM) surrogates



```
...
- The script established deterministic settings:
  - 'global.seed = 42'
  - 'numpy.seed = 42'
  - 'model.random_state = 42'
  - 'cv_shuffle = True'
  - 'cv_random_state = 42'
  - 'lrs.seed = 42'
  - 'surrogate_search.seed.sequence = "42 + round_ids"'

Other reproducibility rules:
- Manifests stored in dir '90-8601' with '2' suffix
- hash raw input dataset bytes with SHA-256 and record it in manifests/logs
- serialized parameters in fixed order:
  - 'cat_id', 'cat_id', 'cat_id', 'cat_id'
  - store physics/logging numbers as JSON numeric values
  - for nonnumeric keys additionally store rounded-to-18-decimal canonical strings

what this step achieved relative to its success criteria:
- bounds explicit and unambiguous for every parameter: yes; exact historical min/max were selected for all 28
  parameters.
- expansion policy numerically/procedurally defined: yes; 5% default cap, 10% hard cap, explicit
  history-based triggers, explicit natural-domain guards, and failure-based values were established.
- logging fields cover evaluation provenance and modeling metadata: yes; required fields for evaluation
  artifacts, manifests, model records, and candidate generation were fully specified.
- reproducible artifact/versioning convention established before modeling: yes; run IDs, round IDs, file
  conventions, hash policy, and deterministic seeds were defined.

Important unresolved points / caveats to remember:
- The conversation did NOT confirm creation of 'step2_bonds_logging.json'; only the script to produce it was
  written.
- No 'generated.json' was created in this step.
- No optimization, surrogate modeling, DDTF design generation, Pareto analysis, or final project artifacts were
  performed here; this step stayed strictly in SCDS.
- The assumed natural domain '[3,10]' for 'max' and '[0.05,1.0]' for the other groups were
  numerically domain-inference choices, not confirmed by tool documentation.
- A default MCP evaluation in this step returned different values from the earlier step's default MCP evaluation;
  later steps should verify whether the tool is stochastic, versioned, or state-dependent.
- The step established serialization and key rules, but no real evaluation log implementation was yet executed
  against them.

Files created/modified during this step:
- Created: 'step2_defining_bonds_logging.py'
- Included output when run later: 'step2_bonds_logging.json' (not confirmed created in transcript)

Best concise handoff for the next executor:
- No 'step2_bonds_logging.json' or the materialize distribution of initial bounds, expansion logic, logging
  schema, artifact conventions, and seed policy.
- Initial optimization bounds should be the exact historical min/max for all 28 parameters listed above.
- Full evaluation should use durations=10, duplicate using 'cat_id' == 'e-9' / 10-decimal canonical rounding.
  Full transient MCP failures at most once, and log all attempts immediately with the required fields.
  All manifests and logs should include the exact run ID.
- Before proceeding, run 'step2_defining_bonds_logging.py' to materialize 'step2_bonds_logging.json' and then
  approximate taken before: 5422s
  approximate taken after: 3702s
...

```

Components in NVIDIA Agent Toolkit

Open stack for building and running enterprise AI agents

OPEN MODELS

Nemotron

Open models for reasoning, safety, and multimodal intelligence.

NIM

Optimized inference microservices that expose models through standard APIs.

Dynamo

Distributed inference orchestration for efficient, scalable model serving.

AGENT TOOLS & SKILLS

NeMo RL & NeMo Gym

Reinforcement learning and scalable environments for post-training, rollouts, and evaluating agent behavior.

NeMo Relay

Control and observability layer for agent runs, tool calls and LLM calls across existing frameworks.

NeMo Switchyard

Task-aware model routing across endpoints to optimize quality, latency and cost.

Omniverse

Libraries for physical AI, simulation, and 3D workflow automation.

BioNeMo

Libraries for life sciences, enabling agents for drug discovery, protein structure prediction, and genomics workflows.

PhysicsNeMo

Open framework for scalable physics AI models for science and engineering.

BLUEPRINTS

NemoClaw

Blueprints for custom, super agents that turn your company's proprietary intelligence, data and domain expertise into action.

AI-Q

Enterprise AI knowledge engine that connects agents to structured and unstructured data – making it understandable, and predictive.

SECURE RUNTIME

OpenShell

Policy-based controls for privacy, networks, tools, files, and code execution.

CUDA-X Agentic Skills for Science

The Foundation of Modern Science

7M

Developers

600M

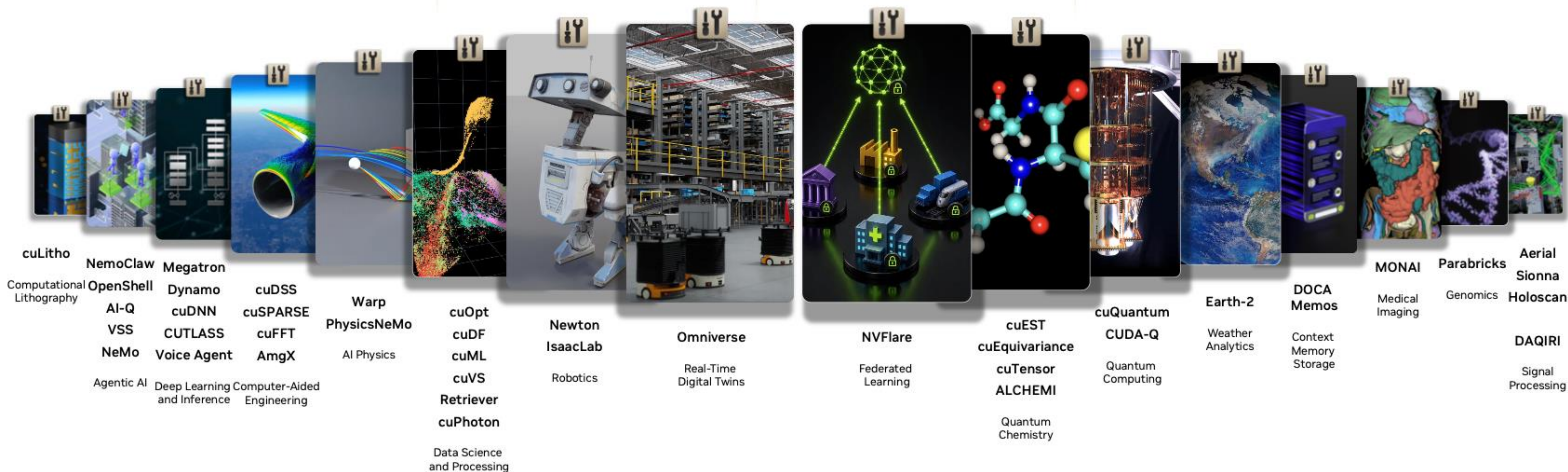
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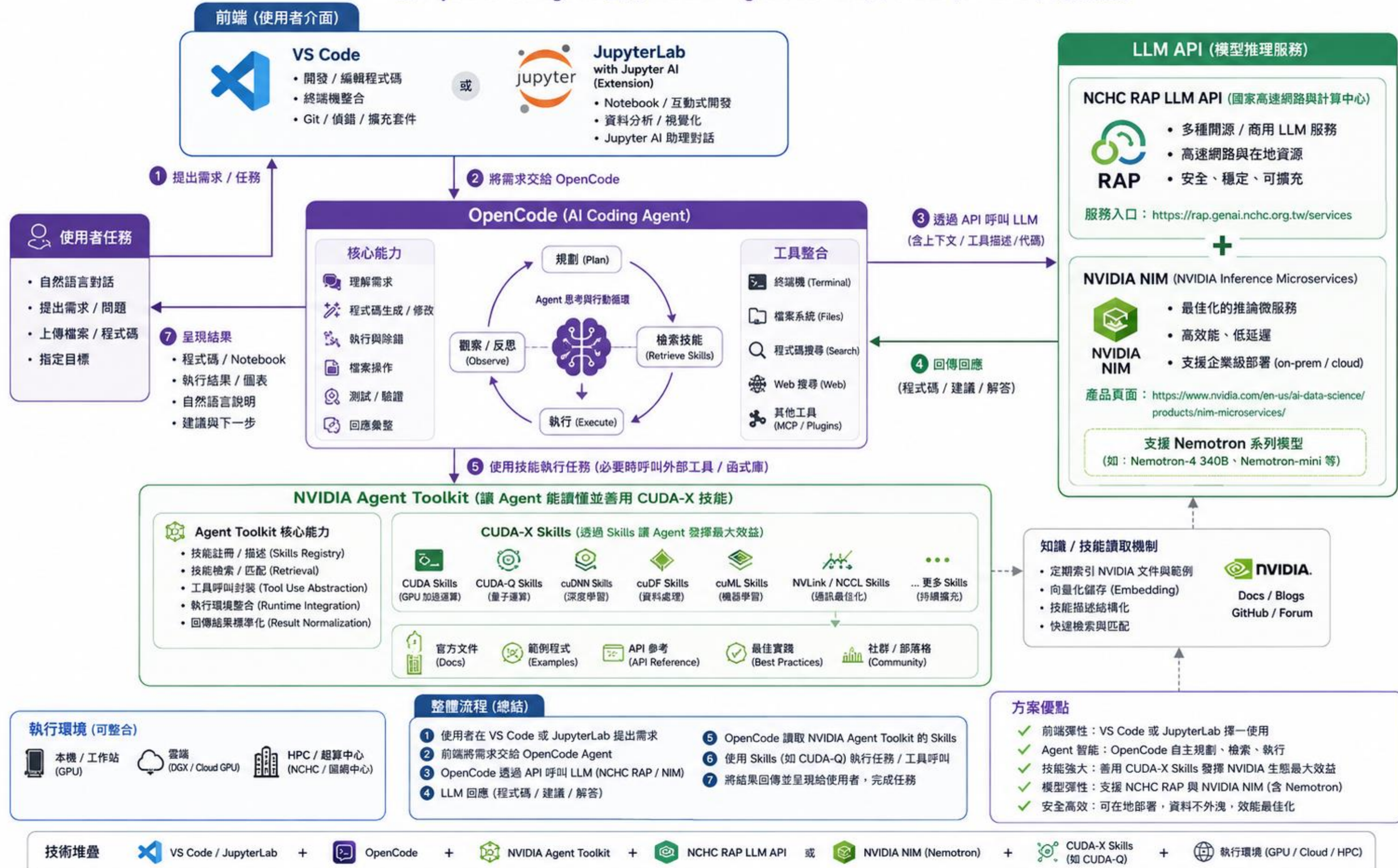
Every Industry | One Architecture | CUDA Everywhere

- + Jupyter AI extension
- + OpenCode
- + RAP LLM API
- + NVIDIA Agent Toolkit Skills

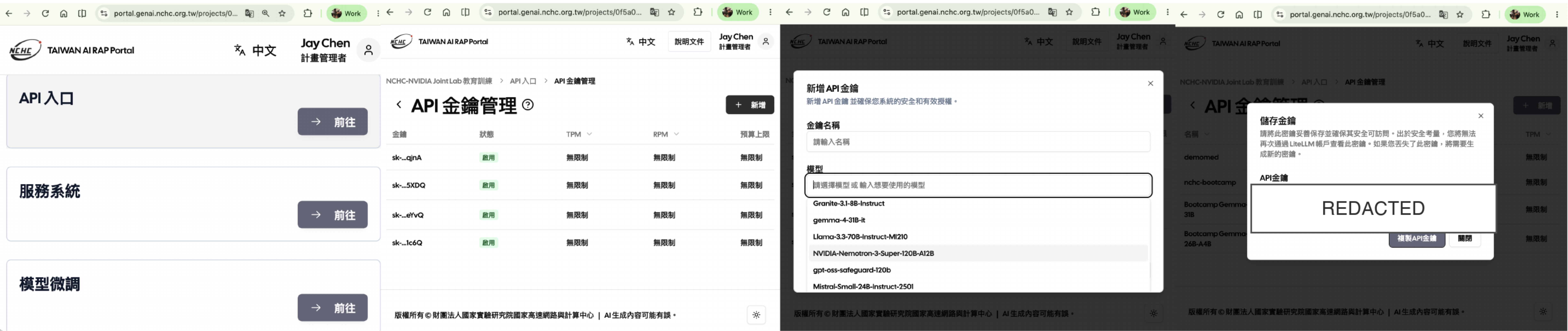
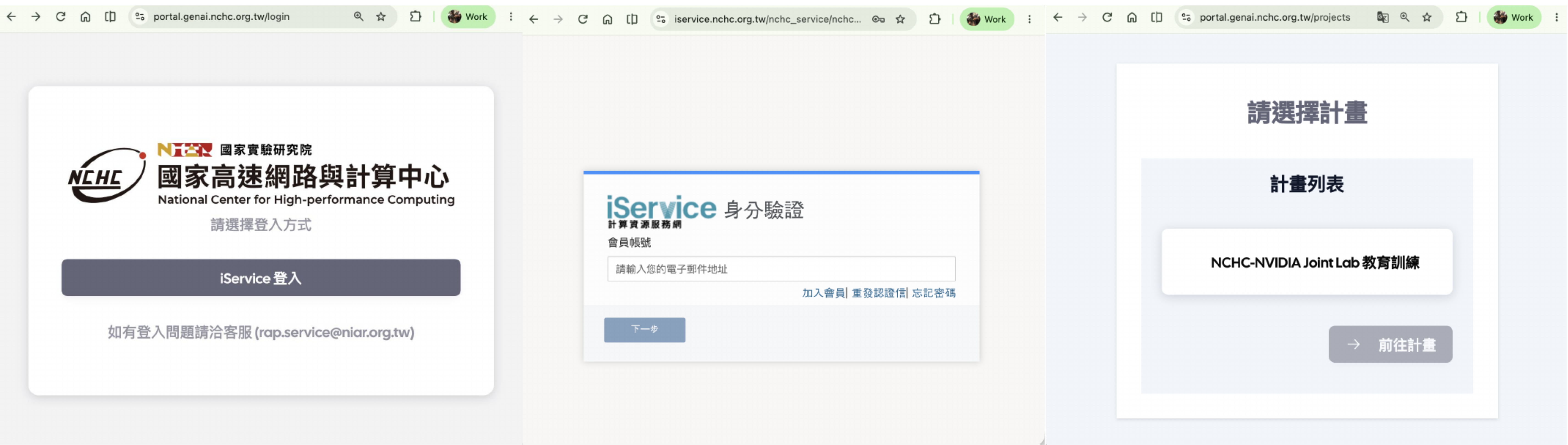


OpenCode + VSCode / JupyterLab + NCHC RAP LLM API / NVIDIA NIM + Nemotron

讓 OpenCode Agent 善用 NVIDIA Agent Toolkit 的 Skills (CUDA-X) 完成任務

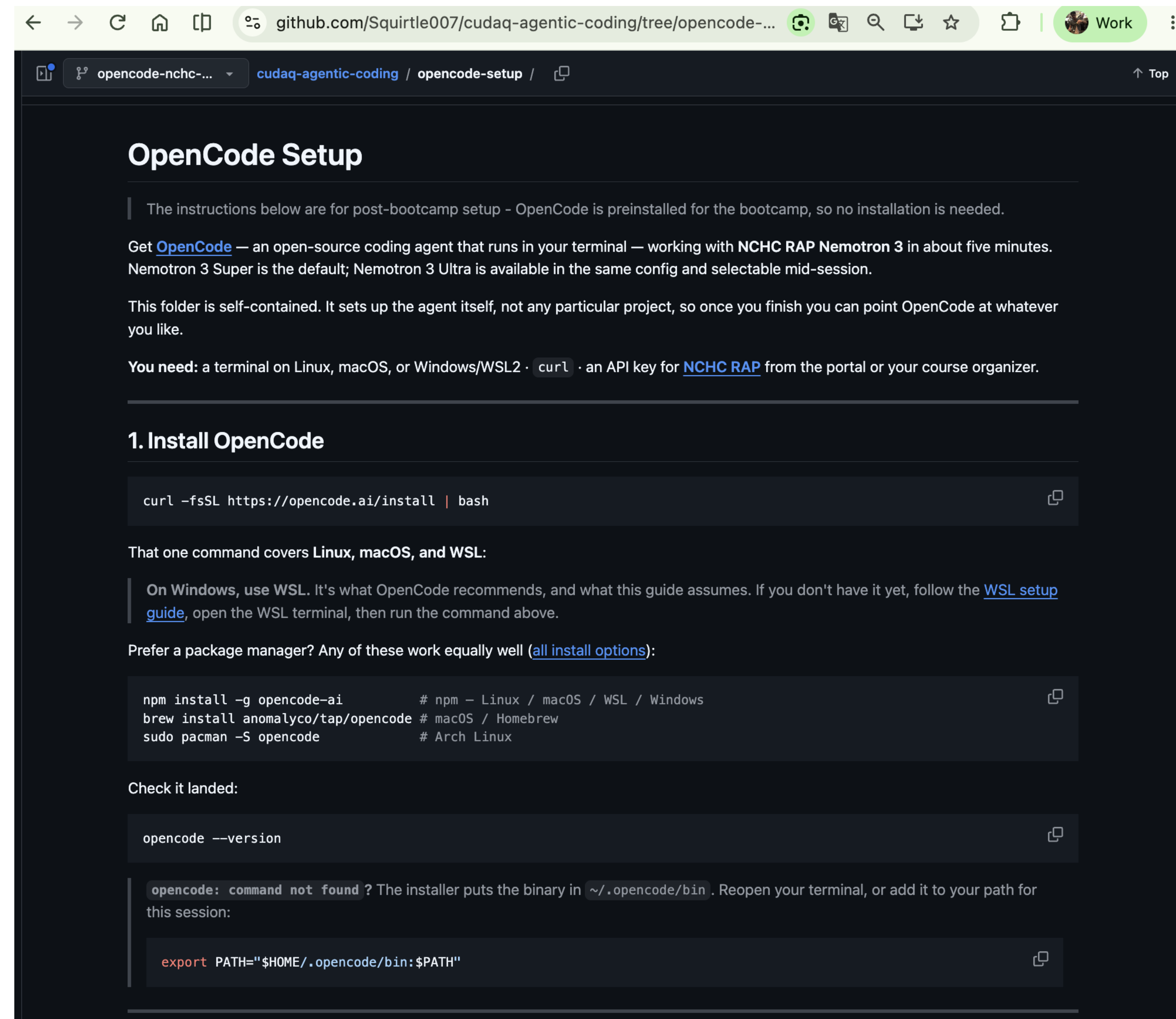


Get a LLM API Key on NCHC RAP



More setup details are open-sourced

<https://github.com/Squirtle007/cudaq-agentic-coding/tree/opencode-nchc-rap/opencode-setup>



← → ↺ 🏠 📄 🔍 ⌵ ☆ 🗂️ | Work ⋮

opencode-nchc-... | cudaq-agentic-coding / opencode-setup | ↑ Top

OpenCode Setup

The instructions below are for post-bootcamp setup - OpenCode is preinstalled for the bootcamp, so no installation is needed.

Get [OpenCode](#) — an open-source coding agent that runs in your terminal — working with **NCHC RAP Nemotron 3** in about five minutes. Nemotron 3 Super is the default; Nemotron 3 Ultra is available in the same config and selectable mid-session.

This folder is self-contained. It sets up the agent itself, not any particular project, so once you finish you can point OpenCode at whatever you like.

You need: a terminal on Linux, macOS, or Windows/WSL2 · `curl` · an API key for [NCHC RAP](#) from the portal or your course organizer.

1. Install OpenCode

```
curl -fsSL https://opencode.ai/install | bash
```

That one command covers **Linux, macOS, and WSL**:

On Windows, use WSL. It's what OpenCode recommends, and what this guide assumes. If you don't have it yet, follow the [WSL setup guide](#), open the WSL terminal, then run the command above.

Prefer a package manager? Any of these work equally well ([all install options](#)):

```
npm install -g opencode-ai      # npm — Linux / macOS / WSL / Windows
brew install anomalyco/tap/opencode # macOS / Homebrew
sudo pacman -S opencode        # Arch Linux
```

Check it landed:

```
opencode --version
```

`opencode: command not found` ? The installer puts the binary in `~/ .opencode/bin` . Reopen your terminal, or add it to your path for this session:

```
export PATH="$HOME/.opencode/bin:$PATH"
```

2026 NCHC Open Hackathon



2026 NCHC Open Hackathon

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Celebrate
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Please ack NCHC and Mentors if any publications.

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3. 機器人模擬資料不夠逼真？MIT 用 3 個 AI Agent 直接「生」出來

4. OpenAI Codex 負責人來台：下一代 AI 的定義，不再只由模型決定

5. 公開人形機器人 25 自由度機器手！1X：手是通往物理世界的 API

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2026 NCHC Open Hackathon

Important Dates

- 09 / 30 (Wed) : Registration deadline
 - <https://www.openhackathons.org/s/siteevent/a0CUP00004AB2oc2AD/se000481>
- 10 / 07 : Connecting Mentors and Teams
 - Slack Channel, Email Group, Line Group ...etc.
- 10 / 14 (Wed) : Day 0: Kick-off Meeting 15:00~17:00 PM (Online)
- 10 / 21 (Wed) : Day 1: Scrum #1 Meeting 15:00~16:00 PM (Online)
- 10 / 28 (Wed) : Day 2: Scrum #2 Meeting 15:00~16:00 PM (Online)
- 11 / 04 (Wed) : Final Day 10:00~16:00 PM (In-person at NCHC)



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Developers

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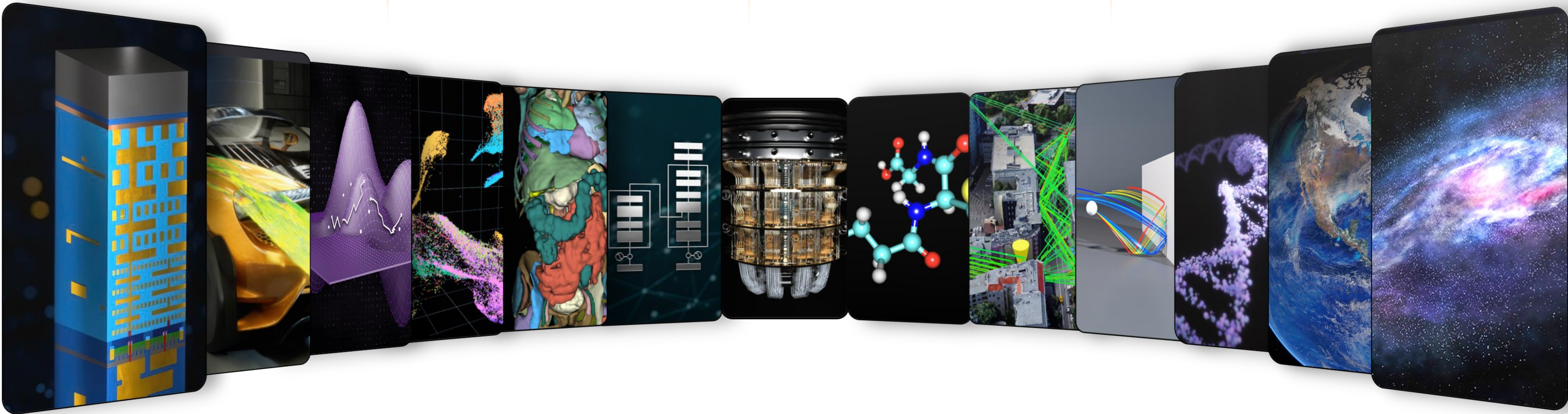
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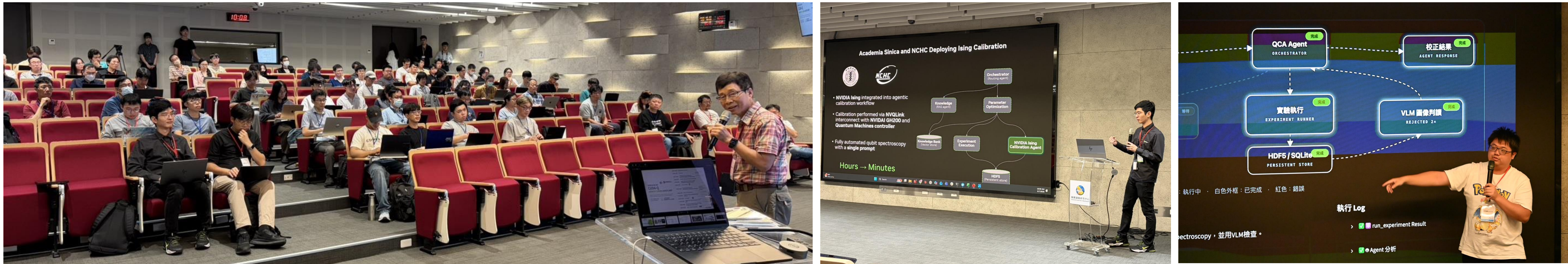
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關鍵議題研究中心1F演講廳

AGENDA

09:30-10:00

Registration & Setup

10:00-10:15

Opening

Academia Sinica | 陳啟東教授

10:15-10:45

NVIDIA Quantum Platform Overview: CUDA-Q & NVIDIA Ising

NVIDIA | Pika Wang

10:45-11:15

NVIDIA CUDA-Q Environment Setup

11:15-12:00

Basic CUDA-Q Hands-on + Ising Calibration Playground (NIM)

12:00-13:00

Lunch Break

13:00-14:00

Superconducting QPU Introduction & Agentic Calibration

Academia Sinica | 陳啟東教授
NCHC | 李泰岳博士, 何明鋼博士

14:00-14:10

NCHC Tech Sharing

NCHC | 李泰岳博士

14:10-14:20

Agentic Coding with NCHC RAP Service

NVIDIA | Jay Chen

14:20-14:40

Tea Break

14:40-15:20

Advanced CUDA-Q Hands-on with RAP

NVIDIA | Pika Wang

15:20-16:00

Bring-Your-Own-Code with CUDA-Q Agentic Programming

16:00-16:20

Q&A + Open Discussion

Q&A

